

How do repeated natural disasters affect local housing price growth and volatility in high-risk counties?

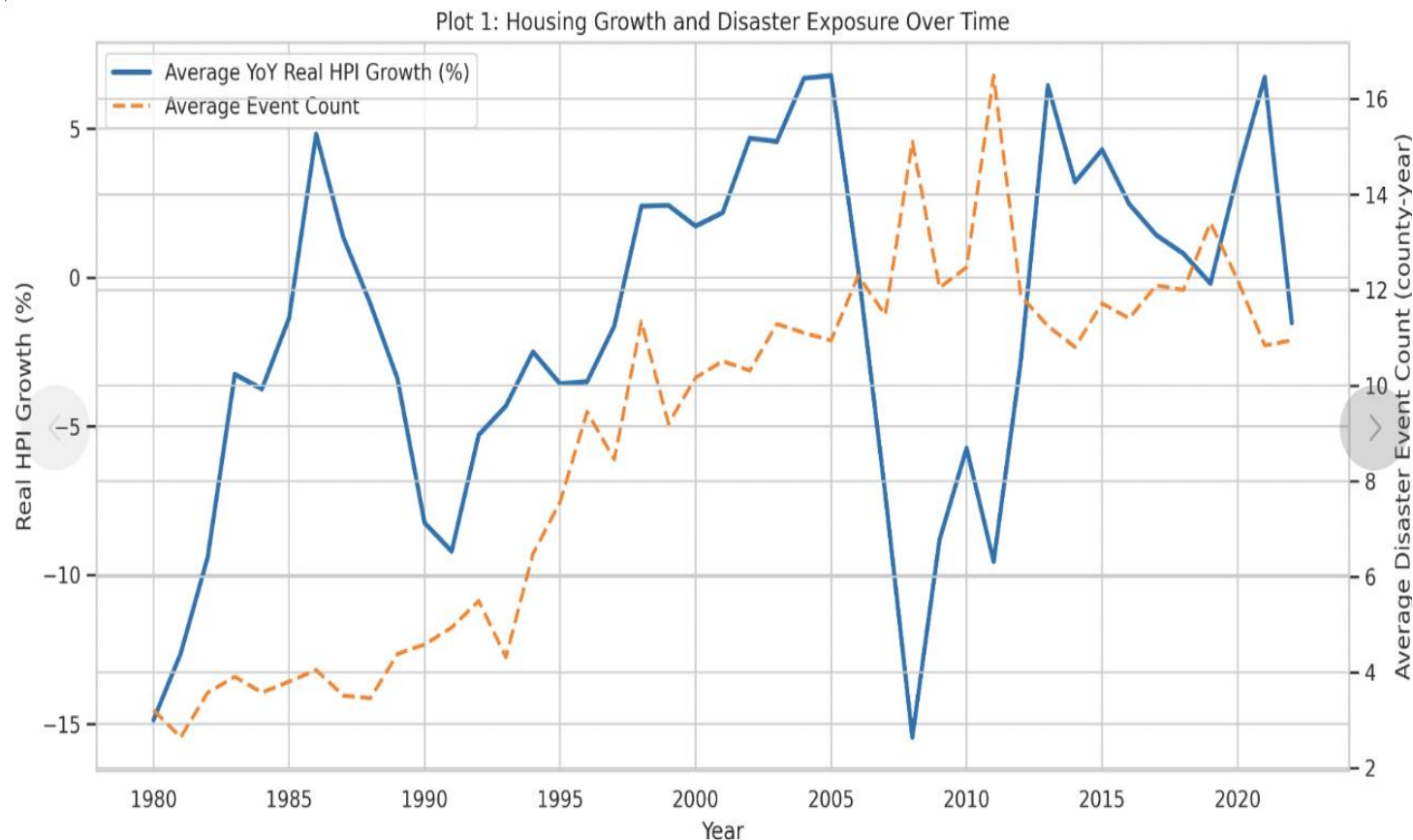
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Context & Data

County panel combining NOAA Storm Events, Yale/Shiller home prices, and FRED macro controls

- 116,137 rows across 3,347 counties, covering 1980-2022
- Two-way fixed effects with county-clustered standard errors
- Goal: show how disaster exposure, housing growth, and macro conditions move together

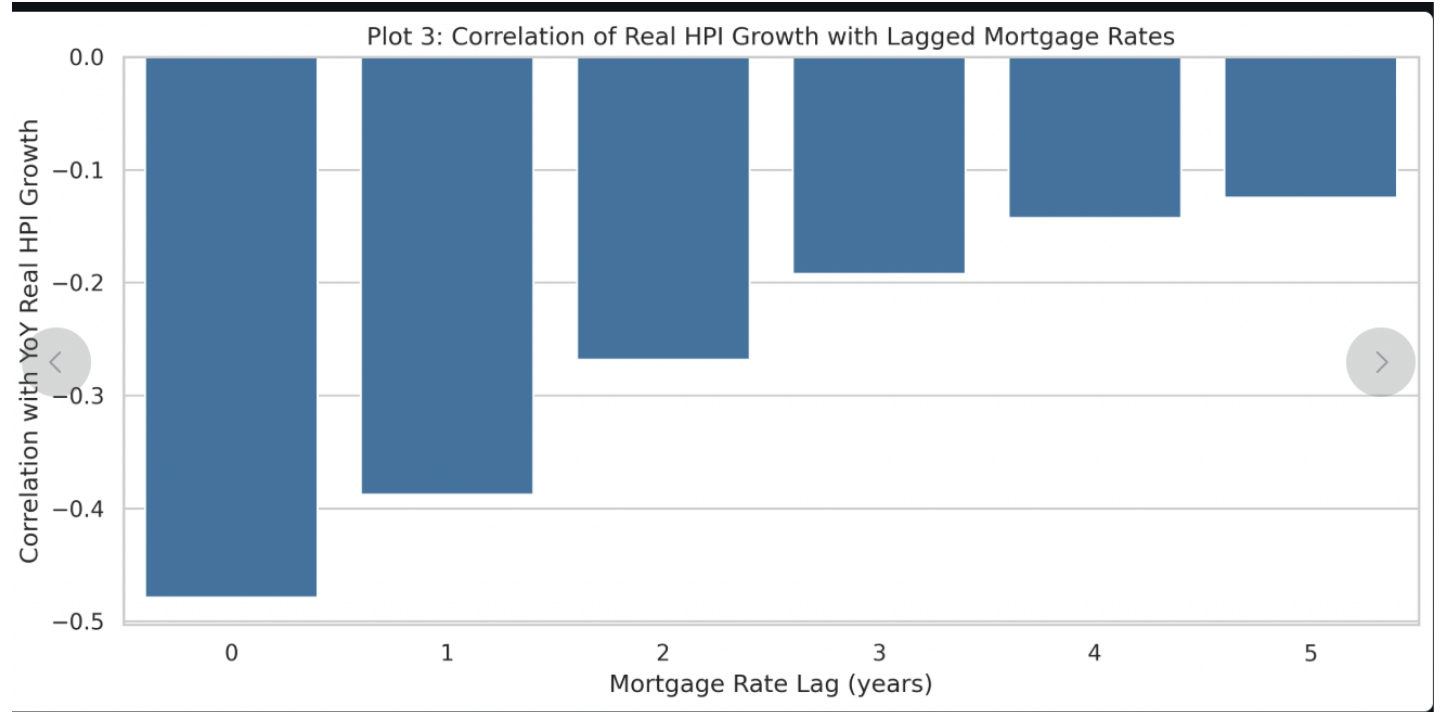
Key Finding



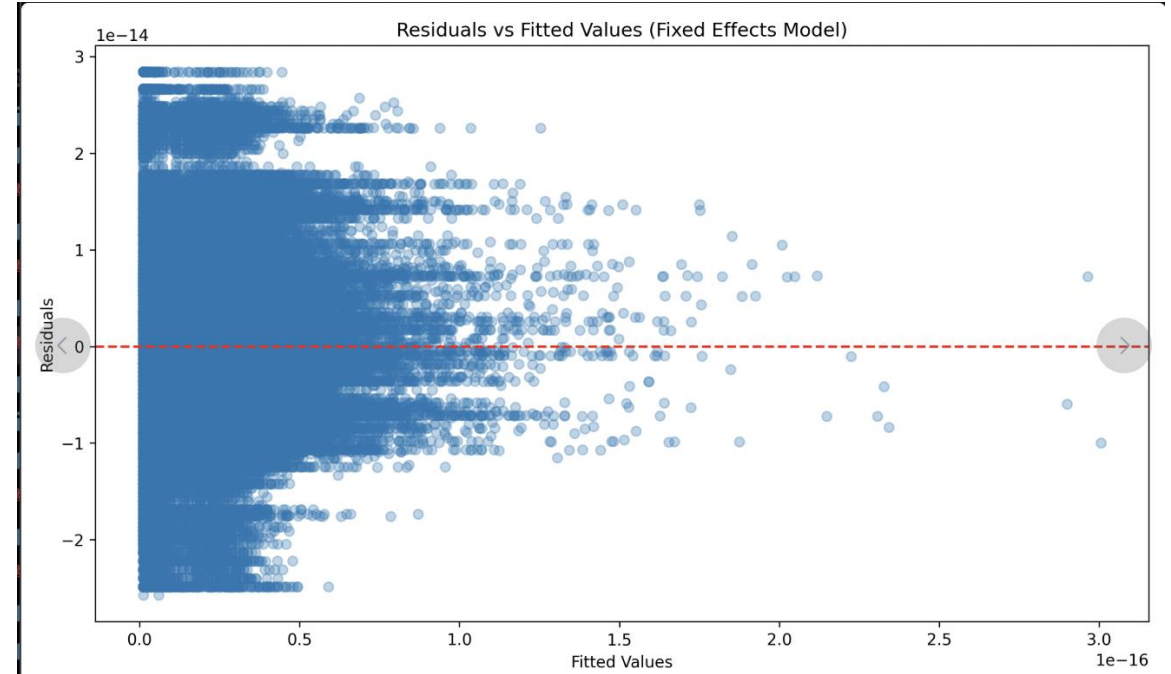
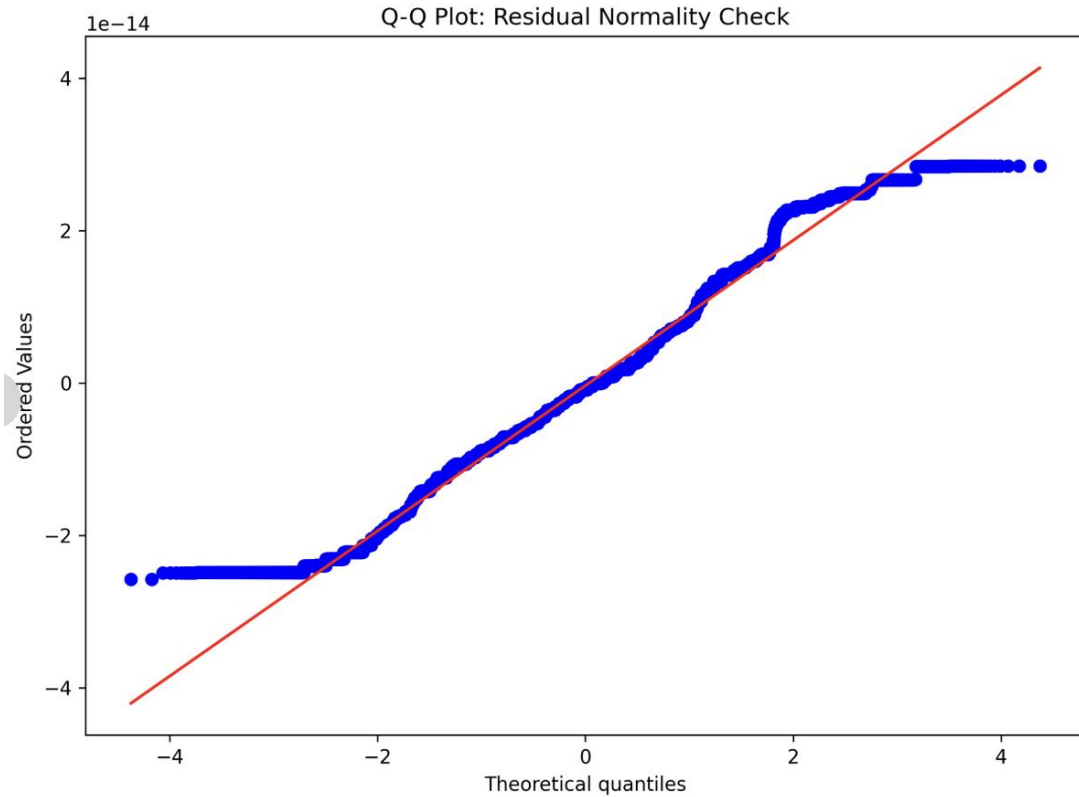
shows how local housing prices and disaster events have changed over time, illustrating that repeated disasters have little effect on overall housing price growth. Outside of the 2008 crisis there may be a weak direct relationship

Main result: Disasters have little effect on housing price growth.

Macro Drivers Dominate



demonstrates that changes in mortgage rates and other macroeconomic factors have a much stronger influence on housing prices than disaster events.

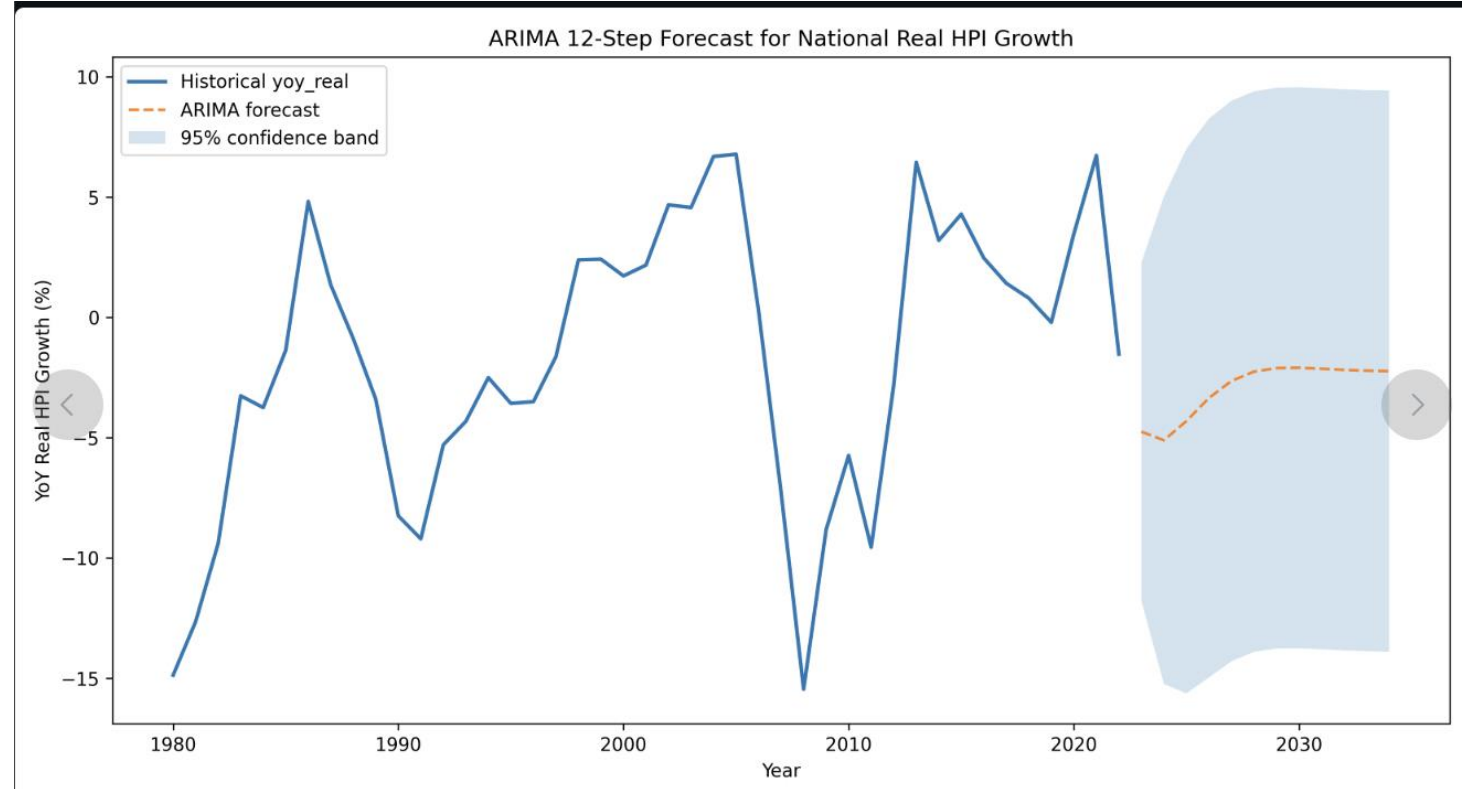


Model Diagnostics

- shows that the model is mostly behaving well: the residuals are scattered around zero without a strong pattern, so there is no obvious misspecification, and the Q-Q plot stays fairly close to the line, meaning the errors are roughly normal with only a few tail deviations.

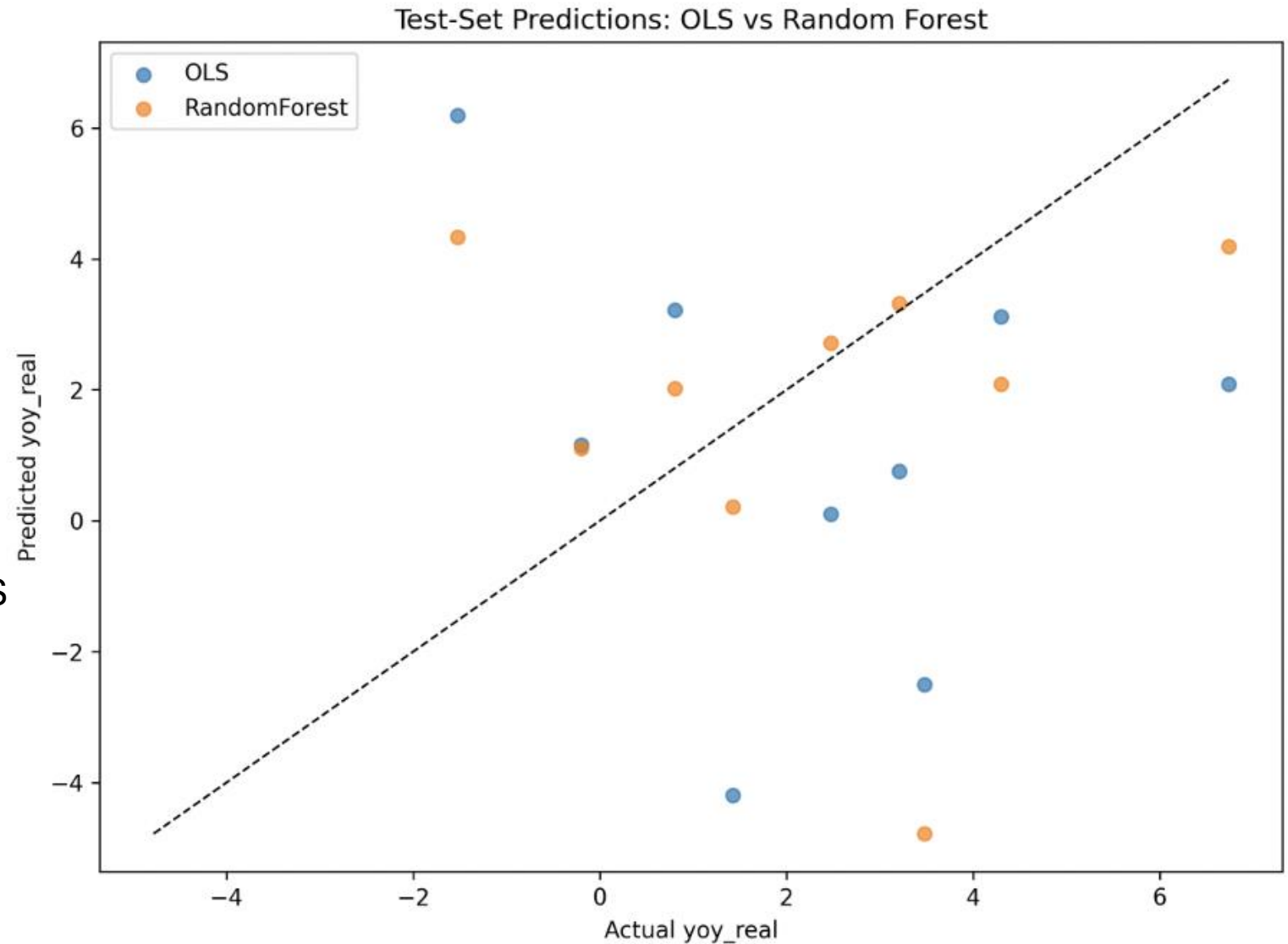
Forecasting Results

- shows the ARIMA forecast compared with the actual housing price series. The lines track each other fairly closely, which means the model captures the overall trend, but the gaps where they diverge show the forecast still misses some short-term movement and error remains.



Machine Learning Comparison

compares the random forest and OLS predictions. The closer line or points mean the model is predicting better, and this visual shows that random forest is only slightly closer to the actual values than OLS, so machine learning improves fit a bit but not dramatically.



Summary & Recommendation

- Disaster history is a secondary risk screen, not the main pricing driver.
- Mortgage rates and broader market fundamentals matter more.
- Investment focus should stay on rate-sensitive markets with strong fundamentals.